

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 214 — Past Questions & Materials
by Renance

Course: CHM 214
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 200

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Paper 1 of 4 — CHM 214 - Structure and Bonding

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

SECOND SEMESTER EXAMINATION FOR 2024/2025 ACADEMIC SESSION

COURSE TITLE: STRUCTURE AND BONDING

UNITS:2

COURSE CODE: CHM 214

TIME ALLOWED: 2 HOURS

ANSWER ANY FOUR

1. (a) What is meant by

(b) Differentiate between

(c) Describe the four

quantum states in an atom?

Explain the difference between an orbital and an orbit.

Define quantum numbers and what they represent?

2. (a) Describe the shapes of the s, p, and d orbitals.

(b) Explain the energy order of orbitals.

(c) Why do electrons occupy orbitals in a specific order and not randomly?

(d) Write the electronic configuration of calcium and explain.

Explain the Aufbau principle.

Is it in a specific order and not randomly?

3. (a) Define bond length,

(b) Define Atomic Spectra

(c) How does

unique?

(d) How does bond angle affect the overall shape of a molecule?

Common bond angles like 109.5°, 120°, and 180°

and list 5 factors that can influence bond length.

Explain the types of atomic spectra you know.

How do atomic spectra help in identifying elements, and what makes an element's spectrum unique?

Explain the value of bond length in a molecule,

Explain the shape of a molecule, and state the geometries of

109.5°, 120°, and 180 degrees.

4. (a). What is the molecule geometry of the following circled elements

be TM, * SD. \ Ge).

N : P

Me ot

oO YY.

(b). What is the molecular geometry of the following: ClF₃, SeBr₄ [PCl₅]. XeF₄, ClF₃

(c). State how lone pairs on central atom affect bond angles in VSEPR theory.

(d). State the main principle behind VSEPR theory, and how it predicts molecular shapes in a molecule, according to

5. (a). Give any four differences between Sigma (σ) bond and Pi (π) bonds

(b) Predict the electronic configuration of Nitric Oxide (NO) and give its bond order

(c) Elucidate the five main points of Valence Bond Theory

(d) Briefly explain s-p overlapping and provide the structure representing the overlap

6. (a). Define a chemical bond. Briefly

(b). Write the Lewis dot formula for

(c). Explain the concept of electro

polarity.

Explain ionic, covalent and dative bond with examples

Write Lewis dot formula for the CO₃²⁻ showing the steps

Electronegativity and its role in determining bond type and

Paper 2 of 4 — CHM 214 - Structure and Bonding

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY

SECOND SEMESTER EXAMINATION FOR 2024/2025 ACADEMIC SESSION

COURSE TITLE: STRUCTURE AND BONDING UNITS:2
COURSE CODE: CHM 214 TIME ALLOWED: 2 HOURS

ANSWER ANY FOUR QUESTIONS

- (a) What is meant by a quantum state in an atom?

(b) Differentiate between an orbital and an orbit. *- orbital part which electrons move in*

(c) Describe the four quantum numbers and what they represent? *→ Principal, Angular momentum, Magnetic, Spin*
- (a) Describe the shapes of the s, p, and d orbitals. *s - spherical, p - dumbbell, d - clove like*

(b) Explain the energy order of orbitals and Aufbau principle. *↑*

(c) Why do electrons occupy orbitals in a specific order and not randomly? *→ because of Hund's rule*

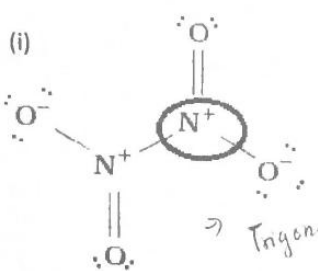
(d) Write the electronic configuration of calcium and explain.
- (a) Define bond length, and list 5 factors that can influence its value in a molecule.

(b) Define Atomic Spectra and list the types you know.

(c) How does atomic spectra help in identifying elements, and what makes elements spectrum unique?

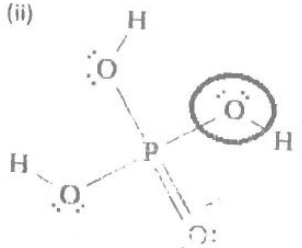
(d) How does bond angle affect the overall shape of a molecule, and state the geometries of common bond angles like 109.5, 120, and 180 degrees.
- (a). What is the molecule geometry of the following circled elements

(i)



→ Trigonal pyramidal

(ii)



Tetrahedral

(b). What is the molecular geometry of the following: ClF_3 , SeF_4 , $[\text{PCl}_5]^-$, XeF_2 , ClF_3

(c). State how lone pairs on central atom affect bond angles in a molecule, according to VSEPR theory.

(d). State the main principle behind VSEPR theory, and how it predict molecular shape

5. (a). Give any four differences between Sigma (σ) bond and Pi (π) bonds
(b) Predict the electronic configuration of Nitric Oxide (NO) and give its bond order
(c) Elucidate the five main points of Valence Bond Theory
(d) Briefly explain s-p overlapping and provide the structure representing the overlap
6. (a). Define a chemical bond. Briefly explain ionic, covalent and dative bond with examples
(b). Write the lewis dot formula or dash formula for the CO_3^{2-} showing the steps
(c). Explain the concept of electronegativity and its role in determining bond type and polarity.

Paper 3 of 4 — CHM 214 ENVIRONMENT CHEMISTRY 2017,2018

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE

COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY f

COURSE TITLE: ENVIRONMENTAL CHEMISTRY CODE: CHM 214

FIRST SEMESTER EXAMINATION (2017/2018 SESSION) TIME: 2 HOURS

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INSTRUCTION: ANSWER ANY THREE QUESTIONS (EACH QUESTION)= 25 marks)

QUESTION ONE

- (a) Differentiate between pollution and pollutants (5marks)
- (b) If Ekpan River suffered a non-point pollution while Agbarho River suffered a point pollution, which of the two is more challenging to remediate and why? (10marks)
- (c) Explain the mechanism of asphyxiation by carbon(ii) oxide (10 marks)

QUESTION TWO

- (a) State the three chemical fluxes in the nitrogen cycle and explain any two of them with the aid of relevant chemical equations. (10marks)
- (b) Explain the three main oxidation reactions of oxygen with earthy materials (10marks) |
- (c) State five effects of acid rain (5marks)

QUESTION THREE

- (a) Mention TWO (2) water quality parameters in each of the following classification: |
 - i. Organoleptic Age p CSET AAS L
 - j, Organics - "Tere. (SORMR 'oysow Cooder Cocker) Tp Coastal Petar
 - iii. Gross Organics Brodhurnrcou mio OLY LW ed CRN) CUreSored otpyen Unr-4 ')
 - iv, nor rganics - Maeqnogaton MMs, Calewert Cs x
 - v. Physico-chemical (10 marks) _ THk Ms < Ss
- _ (2 Biseuss uny Trktt (3) of ihe "Follow: saan te Mangement techniques
- (i) Landfill / © Engineered Landfill (ii) caper (iii) Composting (iv) Pyrolysis and Gasification (20 marks)

o

had?

- (c) ean the following terms
- (i) eutrophication (ji) water quality parameters (5 marks)

QUESTION FOUR

- (a) Explain the effects of greenhouse gases using the experiment you demonstrated? (10 marks)
- (b) Discuss how you would carry out the analysis of any TWO (2) of the following water quality

parameters in a water sample: (i) Turbidity (ii) pH (iii) hardness (iv) salinity (5 marks)

“sums Soap base Kut

(c) Exceeding the standard limits for water quality parameters implies that the water may not be suitable for drinking. Discuss the likely consequences of the following parameters

(i) pH (ii) Salinity (iii) Hardness (iv) Turbidity (v) TDS (5 marks).

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Paper 4 of 4 — CHM 214 ENVIRONMENT CHEMISTRY 2020,2021 Past Questions.

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE
COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

FIRST SEMESTER EXAMINATION 2020/2021 SESSION

NOVEMBER, 2021

COURSE TITLE: ENVIRONMENTAL CHEMISTRY COURSE CODE: CHM214

INSTRUCTION: ANSWER THREE QUESTIONS ONLY

TIME: 2HOURS

QUESTION ONE

- (a) With the aid of chemical equations, explain the formation of acid rain. 8marks
- (b) Differentiate between contamination and pollution. 6marks
- (c) Explain the mechanism for the formation of carboxy-haemoglobin. 9marks

QUESTION TWO

- (a) Briefly explain the environmental effects of greenhouse gases 7marks
- (b) What is the significance of Elementary Cycles? 7marks
- (c) Oxygen is a strong oxidizing agent, and is inherently unstable with many common earth materials. Explain three categories of such reactions. 9marks

QUESTION THREE

- (a) Describe how human activities contribute to increasing greenhouse gases. 8marks
- (b) Describe why the inside of a greenhouse made of glass is "warm". 6marks
- (c) What are the importance and benefits of waste recycling. 9marks

QUESTION FOUR

- (a) Describe the difference between a waste "Dump" and a "Landfill" 8marks
- (b) What is water quality? What does water quality standard means? Name one (international) and two (national) authorities responsible for setting up water quality standards and criteria. 9marks
- (c) Discuss the consequences of having a water body with dissolved oxygen of 2ppm. 6marks

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